Complex Systems and Applications Final Project

Student brief

Choose ONE of the two projects below. Both ask whether information about a network improves prediction of a later event. You will receive a guided base notebook and two small, prepared synthetic CSV files for your choice. Complete the notebook cells in order and write short interpretations in the marked places. The aim is to examine a fair comparison and its limits, not to claim an operational solution.

Each notebook starts with imports and data loading from the course URL. An internet connection is needed for the URL; copies of the CSV files can also be placed beside the notebook for offline work. The notebooks do not generate the synthetic data. A separate construction note explains how each dataset was made.

Option 1 Hospital infection control

A regional hospital group records patient transfers between wards. At the observation point, a few wards already have an active infection cluster. For wards without an active cluster, can transfers recorded during the preceding 14 days help predict a new simulated cluster during the following 14 days?

Central question: Do earlier, directed transfers add predictive information beyond the two ward attributes?

File	Fields and meaning
hospital_wards.csv	ward_id (identifier); occupancy_ratio (occupied-bed fraction); device_use_share (fraction of patients with an invasive device); active_cluster_at_start (known initial status); new_cluster_later (later target).
hospital_transfers.csv	source_ward_id, destination_ward_id, transfer_count. Each row records the direction and volume of transfers before the observation point.

Build a directed network with wards as nodes and recorded transfers as weighted edges. Calculate total incoming transfer volume and incoming transfer volume from wards already affected at the observation point. Compare logistic regression using occupancy ratio and device-use share with the same model after adding these two network features. Only wards without an initial cluster are eligible; the later target is blank for the others.

Interpretation to address: A transfer link does not establish an infection route. A change in model score cannot show that transfers caused clusters or that this model could guide clinical decisions.

Option 2 Power-network disruption

A small transmission network contains substations linked by physical connections. One substation has been interrupted at the observation point. For those still functioning, can their position in the network help predict a simulated interruption over the following 24 hours? The dataset does not simulate a physical cascade.

Central question: Do direct connectivity and local cross-links add predictive information beyond load and equipment age?

File	Fields and meaning
power_substations.csv	substation_id (identifier); load_ratio (pre-disruption load relative to nominal limit); equipment_age_years; initially_interrupted (known initial status); interrupted_later (later target).
power_links.csv	substation_a, substation_b. Each row gives one undirected physical connection already present at the observation point.

Build an undirected network with substations as nodes and physical links as edges. Calculate degree, the number of direct links, and local clustering coefficient, which measures connections among a substation's direct neighbours. Compare logistic regression using load ratio and equipment age with the same model after adding the two network features. Exclude the initially interrupted substation; its later target is blank.

Interpretation to address: These synthetic interruptions do not represent electrical propagation. A better model score, if observed, would not establish a causal process or show that the model is suitable for operating a real grid.

What you must do in either notebook

- Inspect the prepared node and link files, then show later-outcome counts and focused comparisons of the two original attributes among eligible nodes.
- Construct the specified graph from the earlier link file. Calculate and explain the two network features and include a readable network illustration.
- Create one stratified training and test split of the eligible nodes. Fit both logistic regression versions on the same training rows, and evaluate both on the same test rows. Fit scaling on training data only.
- Show accuracy, F1 for the later event, ROC AUC using predicted probabilities, and a confusion matrix for each model. Discuss the errors as well as any difference in scores.
- Write the short answers in the notebook and a 100 to 180 word portfolio summary stating the question, approach, principal result and main limitation.

Submission and portfolio

Submit ONE completed .ipynb file, for the option you chose, with every cell run in order and outputs visible. Put your portfolio summary near the top of the notebook so the completed notebook itself acts as a portfolio item. If you present it through a GitHub repository, a drive folder, a website or another portfolio space, provide a working link or location in your submission. A separate long report is not required. Keep source links or the supplied CSV filenames identifiable, and do not submit a second project.

The synthetic construction notes are provided for transparency. You are not asked to reproduce their generation code or treat synthetic performance as evidence about real hospitals or power systems.